CN LAB 3

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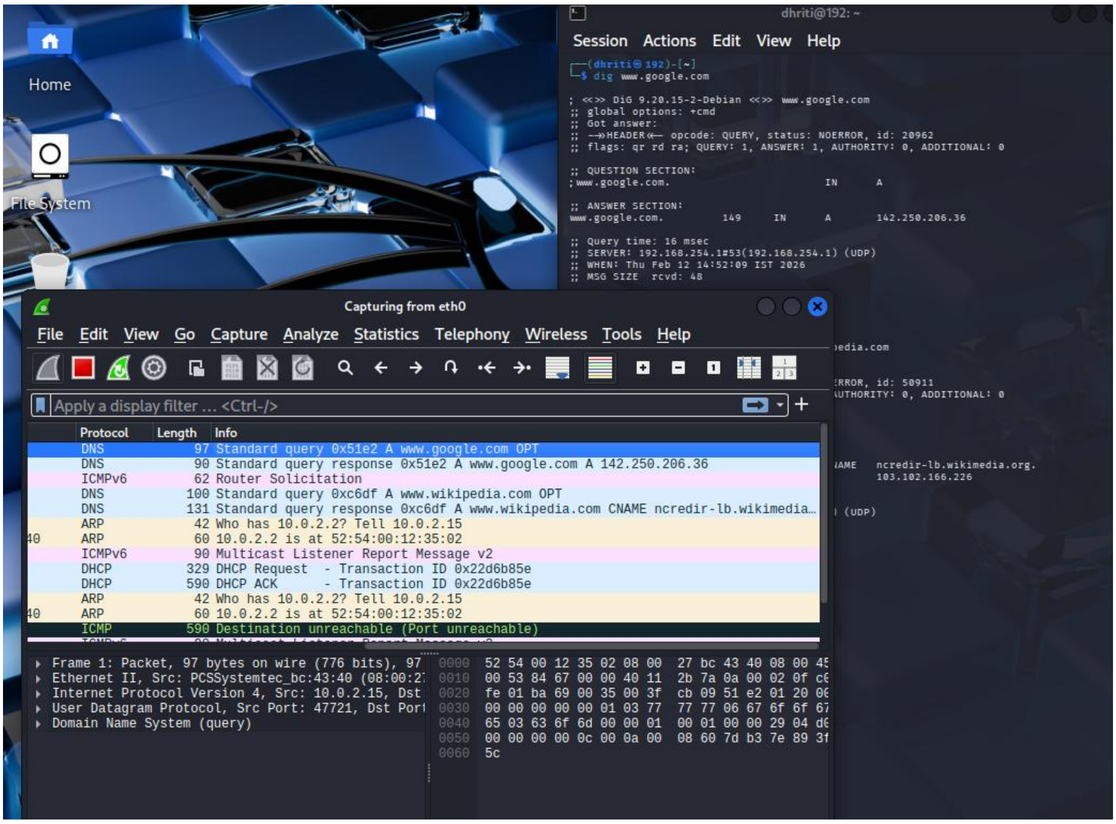
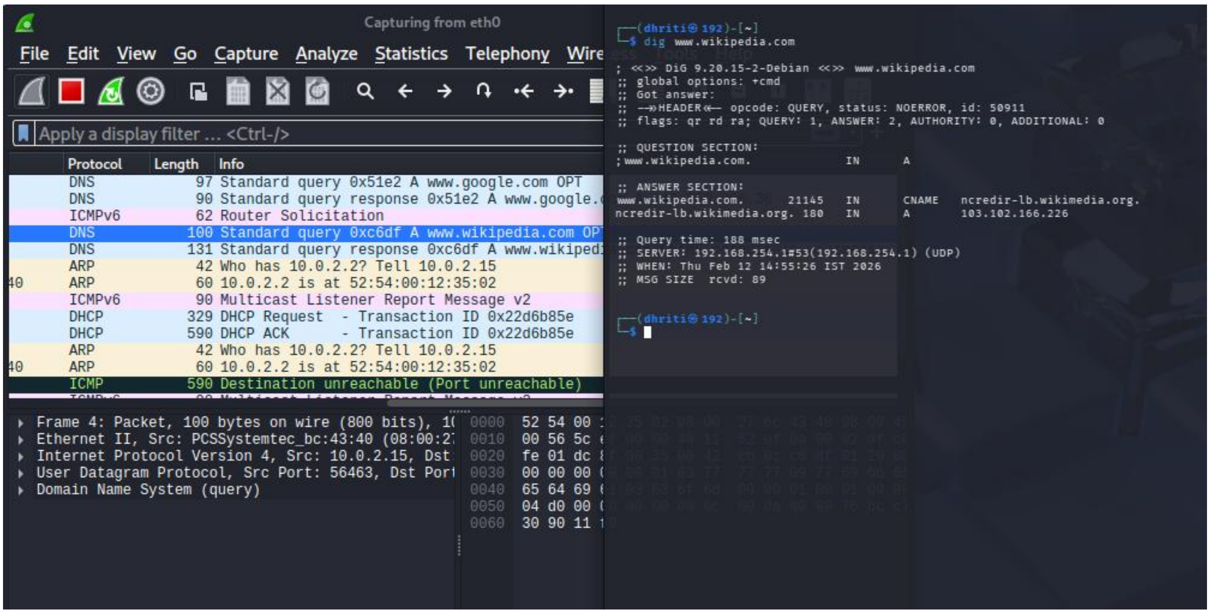
LAB 3.1



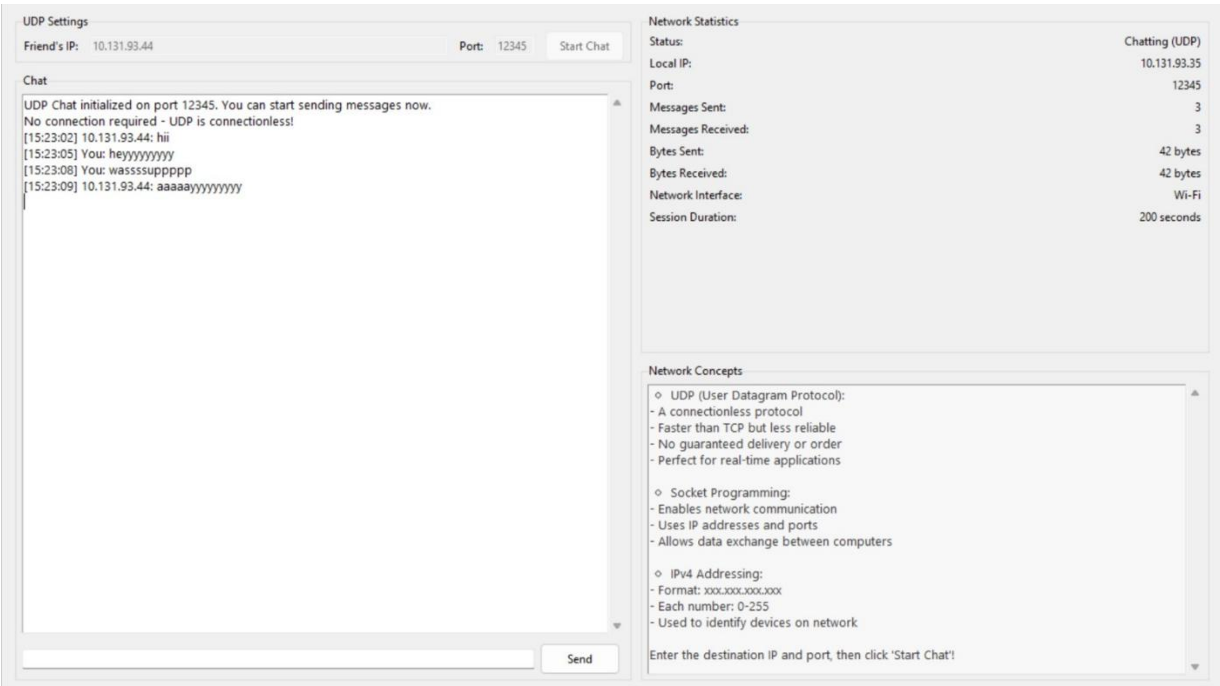


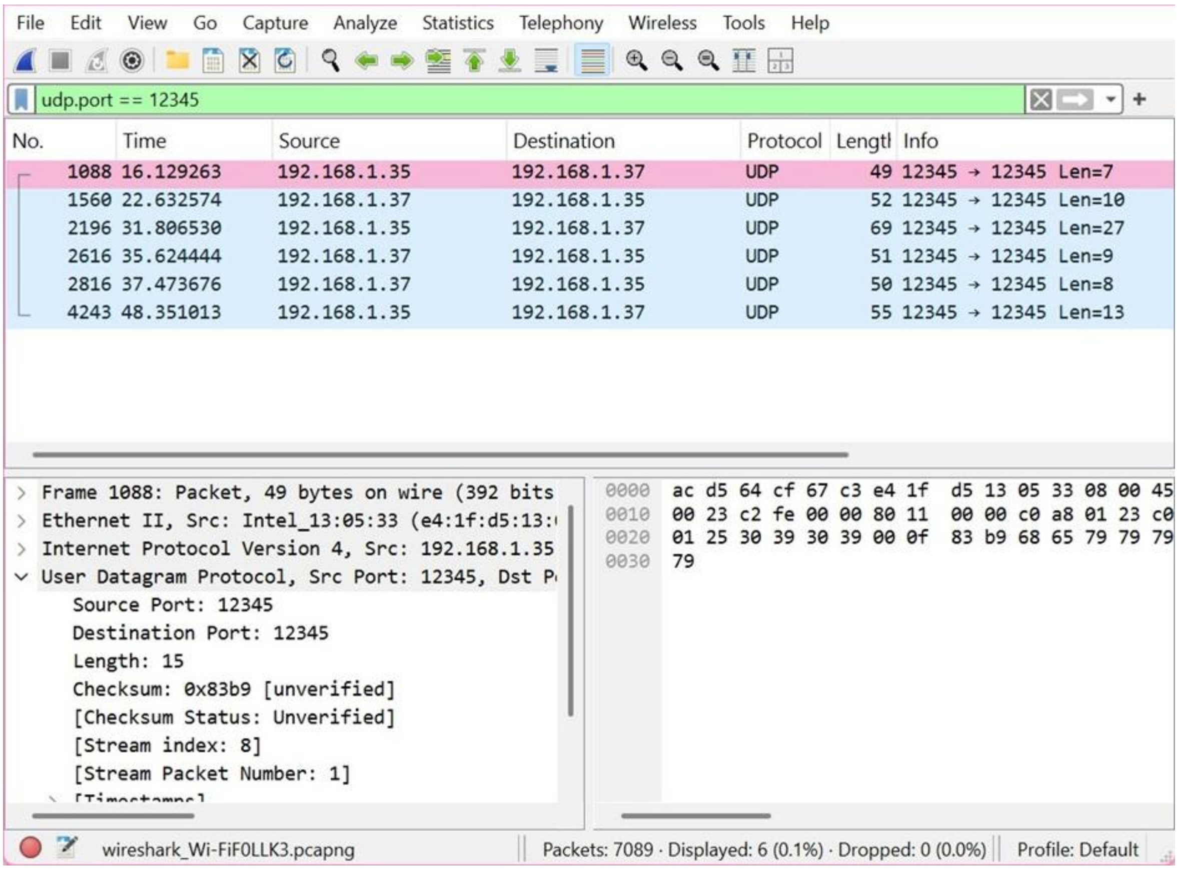
1. **How does UDP differ from TCP?**
   UDP is connectionless and does not use a handshake before sending data.
   TCP is connection-oriented and uses a three-way handshake.
   UDP is faster but does not guarantee delivery.
   TCP is slower but reliable and ensures data is delivered correctly.

2. Why is UDP suitable for DNS?
   DNS queries are small and need fast responses.
   UDP is lightweight and does not require connection setup.
   This makes DNS resolution quick and efficient.
   DNS uses UDP on port 53.

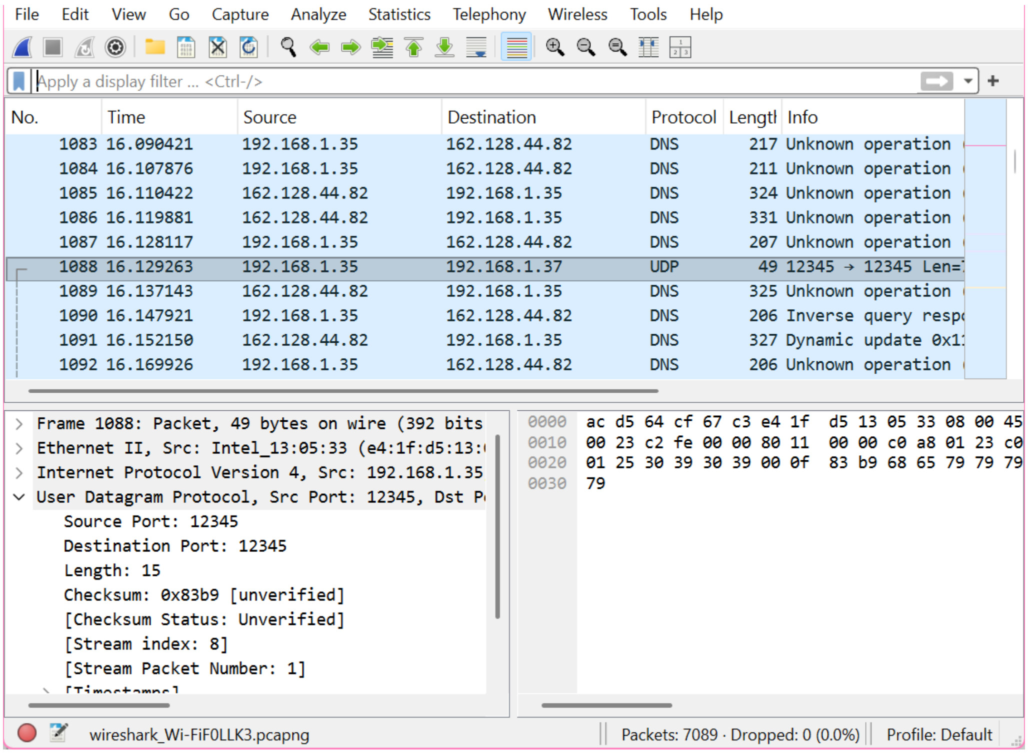
3. Observations about UDP traffic in Wireshark
   UDP packets were seen between two IP addresses.
   Both systems used port 12345.
   No handshake was observed.
   Messages were sent instantly.
   Data was visible inside the UDP packets.

LAB 3.2




File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp.port == 12345
No. Time Source Destination Protocol Lengtl Info
1088 16.129263 192.168.1.35 192.168.1.37 UDP 49 12345 → 12345 Len=7
1560 22.632574 192.168.1.37 192.168.1.35 UDP 52 12345 → 12345 Len=10
2196 31.806530 192.168.1.35 192.168.1.37 UDP 69 12345 → 12345 Len=27
2616 35.624444 192.168.1.37 192.168.1.35 UDP 51 12345 → 12345 Len=9
2816 37.473676 192.168.1.37 192.168.1.35 UDP 50 12345 → 12345 Len=8
4243 48.351013 192.168.1.35 192.168.1.37 UDP 55 12345 → 12345 Len=13
Frame 1088: Packet, 49 bytes on wire (392 bits
Ethernet II, Src: Intel_13:05:33 (e4:1f:d5:13:
Internet Protocol Version 4, Src: 192.168.1.35
User Datagram Protocol, Src Port: 12345, Dst P
Source Port: 12345
Destination Port: 12345
Length: 15
Checksum: 0x83b9 [unverified]
[Checksum Status: Unverified]
[Stream index: 8]
[Stream Packet Number: 1]
0000 ac d5 64 cf 67 c3 e4 1f d5 13 05 33 08 00 45
0010 00 23 c2 fe 00 00 80 11 00 00 c0 a8 01 23 c0
0020 01 25 30 39 30 39 00 0f 83 b9 68 65 79 79 79
0030 79
wireshark_Wi-FiF0LLK3.pcapng
Packets: 7089 · Displayed: 6 (0.1%) · Dropped: 0 (0.0%)
Profile: Default

LAB 3.3


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Lengtl Info
1083 16.090421 192.168.1.35 162.128.44.82 DNS 217 Unknown operation
1084 16.107876 192.168.1.35 162.128.44.82 DNS 211 Unknown operation
1085 16.110422 162.128.44.82 192.168.1.35 DNS 324 Unknown operation
1086 16.119881 162.128.44.82 192.168.1.35 DNS 331 Unknown operation
1087 16.128117 192.168.1.35 162.128.44.82 DNS 207 Unknown operation
1088 16.129263 192.168.1.35 192.168.1.37 UDP 49 12345 → 12345 Len=
1089 16.137143 162.128.44.82 192.168.1.35 DNS 325 Unknown operation
1090 16.147921 192.168.1.35 162.128.44.82 DNS 206 Inverse query resp
1091 16.152150 162.128.44.82 192.168.1.35 DNS 327 Dynamic update 0x1
1092 16.169926 192.168.1.35 162.128.44.82 DNS 206 Unknown operation
Frame 1088: Packet, 49 bytes on wire (392 bits
Ethernet II, Src: Intel_13:05:33 (e4:1f:d5:13:
Internet Protocol Version 4, Src: 192.168.1.35
User Datagram Protocol, Src Port: 12345, Dst P
Source Port: 12345
Destination Port: 12345
Length: 15
Checksum: 0x83b9 [unverified]
[Checksum Status: Unverified]
[Stream index: 8]
[Stream Packet Number: 1]
0000 ac d5 64 cf 67 c3 e4 1f d5 13 05 33 08 00 45
0010 00 23 c2 fe 00 00 80 11 00 00 c0 a8 01 23 c0
0020 01 25 30 39 30 39 00 0f 83 b9 68 65 79 79 79
0030 79
wireshark_Wi-FiF0LLK3.pcapng
Packets: 7089 · Dropped: 0 (0.0%)
Profile: Default